Elasticity

→ Elasticity is tendency of a body to resist any change in its shape, size and geometry.

→ If external deforming force is removed, then body have tendency to regain its actual shape, size and geometry.

Perfectly elastic body: It attains the exact actual shape, size and geometry when external force is removed.

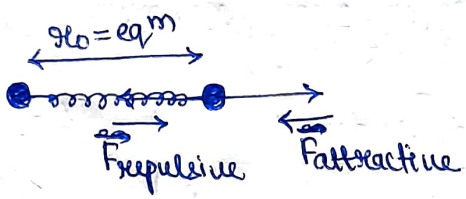
Perfectly plastic body: They have no tendency to regain actual shape, size and geometry when external force is removed.

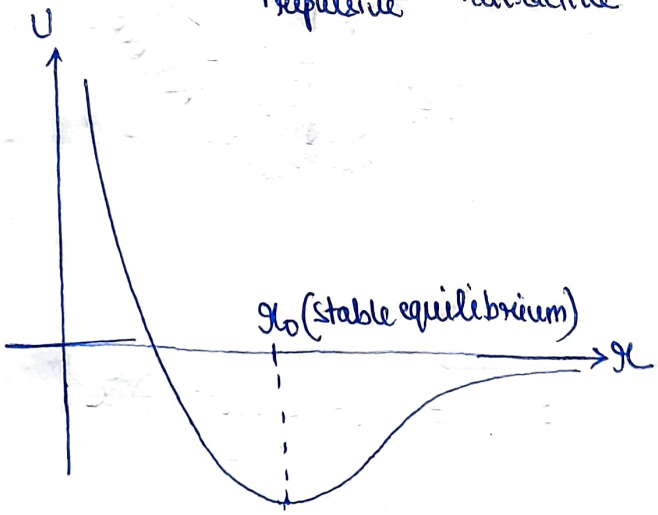
Elasticity ⇒ solids > liquids > gases

* Steel is more elastic than rubber

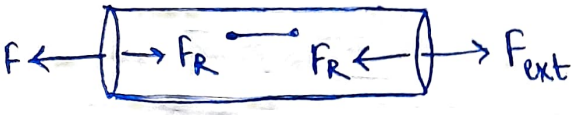
→ Elasticity is due to intermolecular forces

→ If Temp↑ ⇒ Sep. b/w molecules↑ ⇒ Intermolecular force↓ ⇒ Elasticity↓





*



F_{ext} = External deforming force

F_R = Internal reforming force

→ If system is at equilibrium then $\boxed{F_{ext} = F_R}$

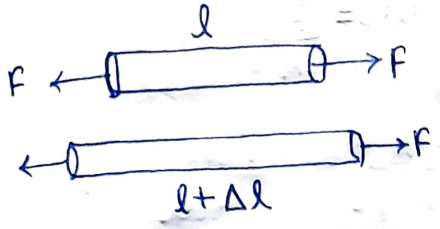
Strain: Fractional change in configuration due to external deforming force

$$\text{Strain} = \frac{\text{Change in configuration}}{\text{Actual Configuration}}$$

→ Unitless
→ Dimensionless

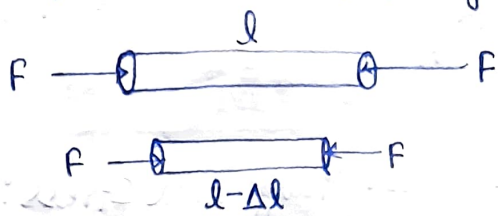
(i) Longitudinal Strain:

a) Tensile strain {If length increases}



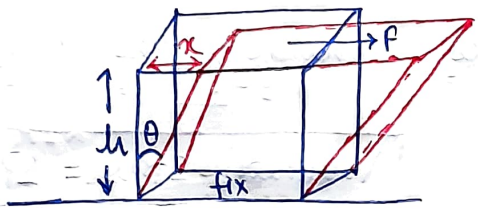
$$\text{Tensile Strain} = \frac{\Delta l}{l}$$

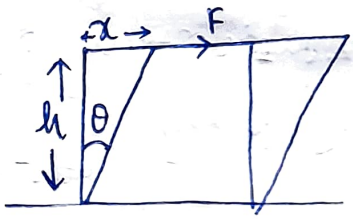
b) Compressive Strain (If length decrease



$$\text{Compressive Strain} = \frac{\Delta l}{l}$$

(ii) Shear Strain/Tangential Strain:



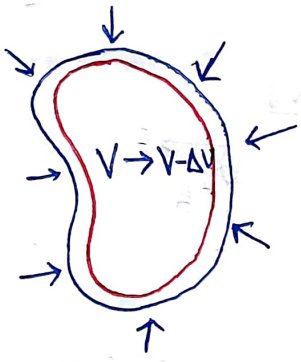


$$\text{Shear Strain} = \frac{x}{h} = \tan\theta$$

Generally, $\theta \approx$ small $\Rightarrow \tan\theta \approx \theta$

$\theta \rightarrow$ Radians

(iii) Volumetric Strain:



$$\text{Volumetric Strain} = \frac{\Delta V}{V}$$

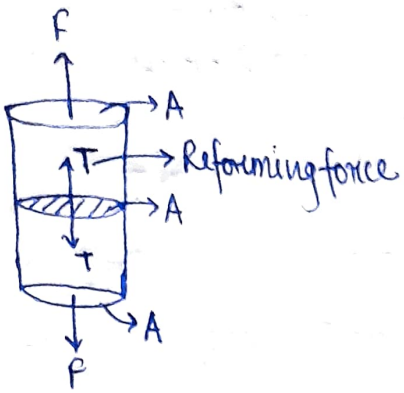
Stress: It is a physical quantity that defines force per unit area.

→ Unit = N/m^2

→ Dimension = $\frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

→ Scalar Quantity

(i) Longitudinal Stress / Normal Stress :

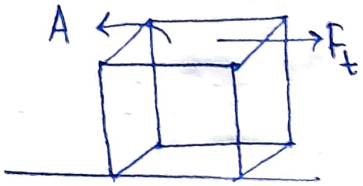


$$\text{Longitudinal Stress} = \frac{\text{Reforming stress}}{\text{Cross-Sectional Area}} = \frac{T}{A}$$

In general, system at eqm $\Rightarrow T = F$

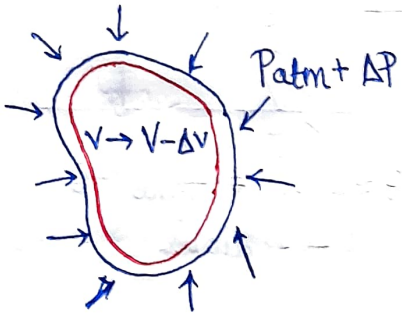
$\Rightarrow$ Longitudinal Stress $= \frac{F}{A}$

(ii) Shear Stress / Tangential Stress :



$$\text{Shear Stress} = \frac{F_t}{A}$$

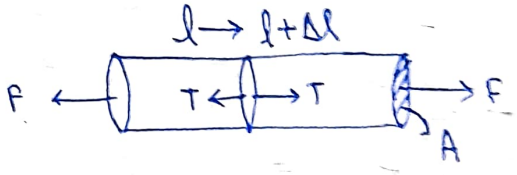
(iii) Volumetric Stress : Pressure or change in pressure



$$\text{Volumetric Stress} = \Delta P$$

Young's Modulus :

$$Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}}$$

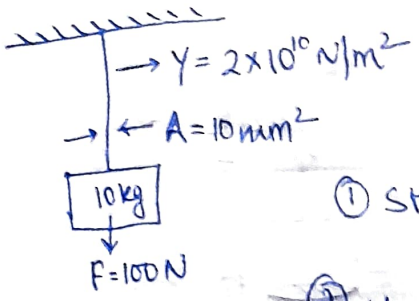


$$Y = \frac{T/A}{\Delta l / l} = \frac{Tl}{A\Delta l}$$

$$T = \frac{YA\,\Delta l}{l}$$

- Property of matter
- If $Y \uparrow \Rightarrow$ Elasticity $\uparrow$
- Unit $= N/m^2$
- Dimension : $[ML^{-1}T^{-2}]$
- Scalar

Q



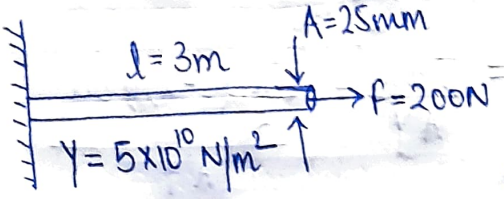
Find ① Stress in wire ② strain in wire ③ Change in length of wire due to block of mass 10kg if actual length is 2m

① $\text{Stress} = \frac{T}{A} = \frac{100\,N}{10\times10^{-6}\,m^2} = 10^7\ N/m^2$

② $Y = \frac{\text{stress}}{\text{strain}} \Rightarrow \text{Strain} = \frac{\text{Stress}}{Y} = \frac{10^7}{2\times10^{10}} = 5\times10^{-4}$

③ $\text{stress} = \frac{\Delta l}{l} \Rightarrow \Delta l = l \times \text{stress} = 2\times5\times10^{-4} = 10^{-3}\,m = 1\,mm$

Q



Find:

① $\text{Stress} = \frac{T}{A} = \frac{200\,N}{25\times10^{-6}\,m^2} = 8\times10^6\ N/m^2$

② $\text{strain} = \frac{\text{stress}}{Y} = \frac{8\times10^6}{5\times10^{10}} = 1.6\times10^{-4}$

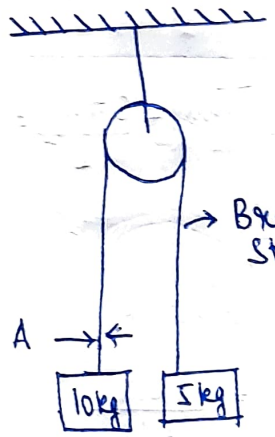
③ Change in length:

$\Delta l = \text{Strain} \times l$
$= 1.6\times10^{-4}\times3$
$= 4.8\times10^{-4}\,m$

④ Final length:

$= 3m + 4.8\times10^{-4}\,m$

Q



Find cross-sectional area of string so that it doesn't break.

$\Rightarrow \text{Stress} = \frac{T}{A} = \frac{200\,N}{3A}$

Now, $\frac{200\,N}{3A} \le 5\times10^6\ \frac{N}{m^2}$

$= \frac{200}{3\times5\times10^6} \le A$

$A \ge \frac{40}{3}\times10^{-6}\,m^2$

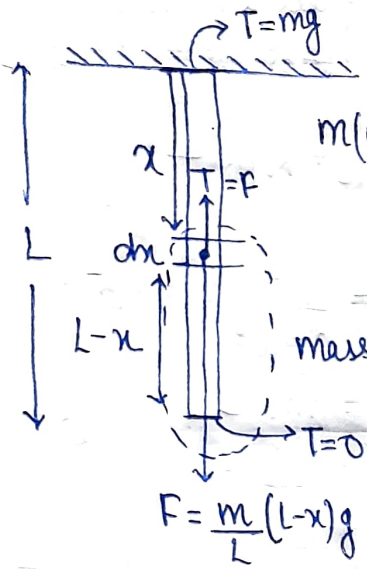
$T = \frac{2m_1m_2g}{m_1+m_2}$

$T = \frac{2\times10\times5\times10}{15_3} = \frac{200\,N}{3}$

$\boxed{A \ge \frac{40}{3}\,mm^2}$

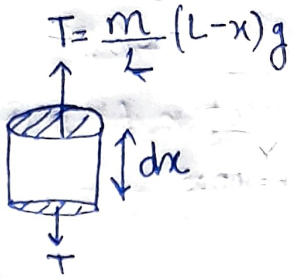
$\boxed{A_{min} = \frac{40}{3}\,mm^2}$

Elongation due to self-weight:



m (uniformly distributed)
A, L, Y

mass = $\frac{m}{L}(L-x)$



Let increase in length be dl.

$$Y = \frac{T/A}{dl/dx} = \frac{T\,dx}{A\,dl}$$

$$dl = \frac{T\,dx}{AY} = \frac{\frac{m}{L}(L-x)g\,dx}{AY}$$

$$\Delta l = \int dl = \frac{mg}{LAY}\int_0^L (L-x)\,dx$$

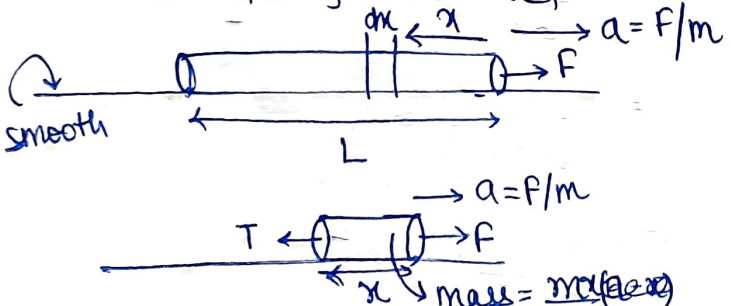
$$\Delta l = \frac{mg}{LAY}\left(Lx - \frac{x^2}{2}\right)_0^L = \frac{mg}{LAY}\left(L^2 - \frac{L^2}{2}\right) = \frac{mgL^2}{2LAY}$$

$$\boxed{\Delta l = \frac{mgL}{2AY}}$$ → Elongation due to self-weight

Trick: If tension T varies linearly with position x, $T_{avg} = \frac{0+mg}{2} = \frac{mg}{2}$

$$Y = \frac{T_{avg}/A}{\Delta l/l} = \frac{T_{avg}L}{A\Delta l} \Rightarrow \Delta l = \frac{T_{avg}L}{AY} = \frac{mgL}{2AY}$$

Q. Find elongation in rod (A, L, Y)
m = uniformly distributed



$$F - T = \frac{mx}{L} \times \frac{F}{mx}$$

$$T = F - \frac{Fx}{L} \Rightarrow \boxed{T = F\left(1 - \frac{x}{L}\right)}$$

$$Y = \frac{T/A}{dl/dx} = \frac{T\,dx}{A\,dl}$$

$$dl = \frac{T\,dx}{AY} = \frac{F\left(1-\frac{x}{L}\right)dx}{AY}$$

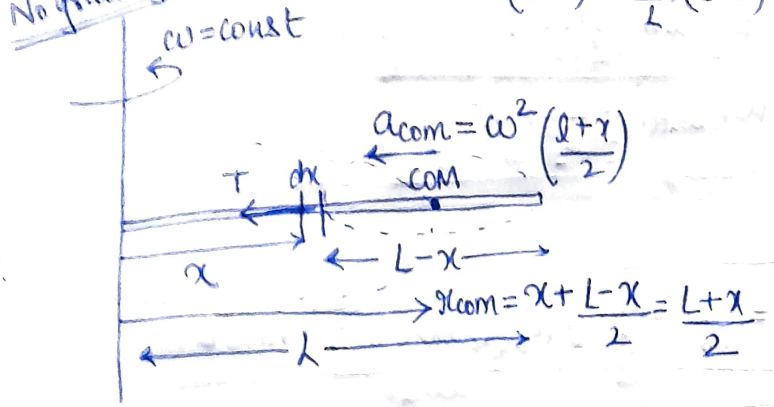
$$\Delta l = \int dl = \frac{F}{AY}\int_0^L \left(1 - \frac{x}{L}\right)dx = \frac{F}{AY}\left(x - \frac{x^2}{2L}\right)$$

$$\Delta l = \frac{F}{AY}\left(L - \frac{L^2}{2L}\right) = \frac{F}{AY}\left(L - \frac{L}{2}\right)$$

$$\boxed{\Delta l = \frac{PL}{2AY}}$$

Q Find elongation of rod (m, l, A, Y)

No gravity



mass in $(L-x) = \frac{m}{L}(L-x)$

$F_{ext} = ma_{com}$

$T = \frac{m}{L}(L-x)\,\omega^2\left(\frac{L+x}{2}\right)$

$$\boxed{T = \frac{m\omega^2}{2L}(L^2 - x^2)}$$

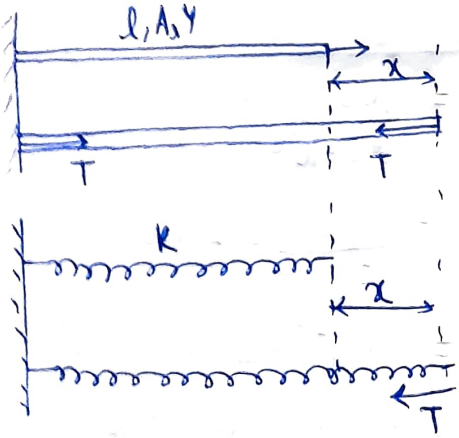
Now, $Y = \frac{T\,dx}{A\,dl}$

$\Rightarrow dl = \frac{T\,dx}{AY}$

$$\Delta l = \int dl = \frac{m\omega^2}{2LAY}\int_0^L (L^2 - x^2)\,dx = \frac{m\omega^2}{2LAY}\left(L^3 - \frac{L^3}{3}\right) = \frac{m\omega^2}{2LAY} \times \frac{2L^3}{3} = \frac{m\omega^2 L^2}{3AY}$$

$$\therefore \boxed{\Delta l = \frac{m\omega^2 L^2}{3AY}}$$

Analogy b/w rod and spring:



$Y = \frac{T/A}{x/l} = \frac{Tl}{Ax} \Rightarrow \boxed{T = \frac{YA}{l}x}$ — (1)

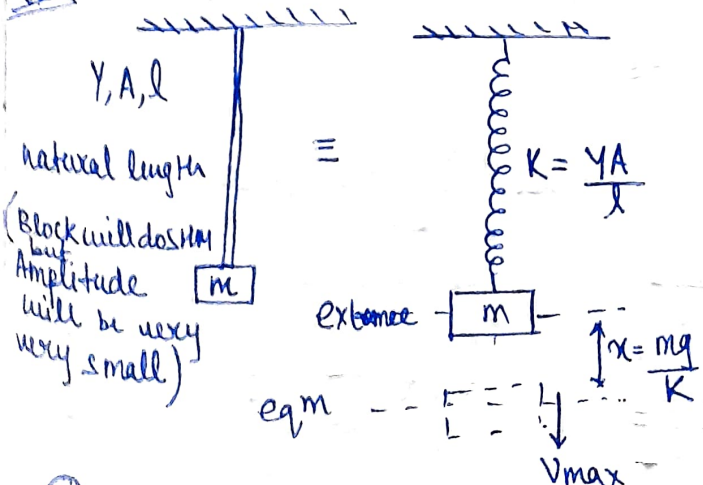
$\boxed{T = kx}$ — (2)

Comparing (1) and (2)

$K = \frac{YA}{l}$

∴ Rod can be replaced with spring with spring constant $K = YA/l$.

Ex.



Find: ① Amplitude (A')

$A' = \frac{mg}{K} = \frac{mgl}{YA}$

② $\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{YA}{lm}}$

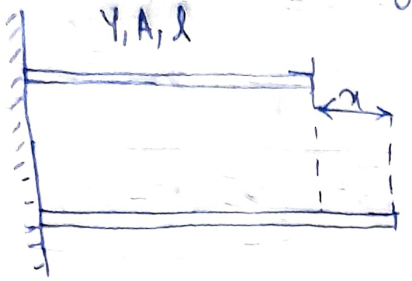
③ $V_{max} = \omega A' = \sqrt{\frac{YA}{lm}} \times \frac{mgl}{YA}$

④ Energy of SHM $= \frac{1}{2}mV_{max}^2$

$= \frac{1}{2}m\left(\frac{YA}{lm} \times \frac{m^2g^2l^2}{Y^2A^2}\right) = \frac{m^2g^2l}{2YA}$

⑤ $T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{ml}{AY}}$

Elastic Potential Energy:



Spring potential energy $= \frac{1}{2}kx^2$

By comparison, we say

$$U_{elastic} = \frac{1}{2}kx^2 = \frac{1}{2}\frac{YA}{l}x^2$$

Now,

$$U = \frac{1}{2}Y\left(\frac{x}{l}\right)(Ax) = \frac{1}{2}Y(\text{strain})(\text{Change in Volume } \Delta V)$$

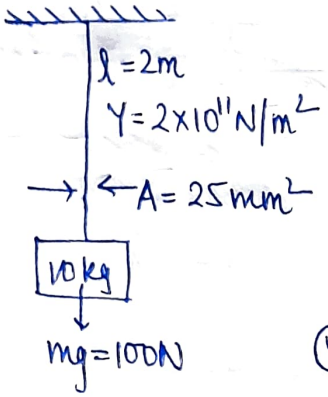
or

$$U = \frac{1}{2}Y(Al)\left(\frac{x}{l}\right)^2 = \frac{1}{2}Y(v)(\text{strain})^2 = \frac{1}{2}Y(\text{strain})^2(\text{Volume})$$

Also, $\star$ $U = \frac{1}{2}(\text{stress})(\text{strain})(\text{Volume})$

Elastic Potential energy density $= \frac{U}{\text{Vol.}} = \frac{1}{2}Y(\text{strain})^2 = \frac{1}{2}(\text{stress})(\text{strain})$

Q



Find: ① Stress $= \frac{T}{A} = \frac{100}{25\times10^{-6}} = 4\times10^6\ N/m^2$

② Strain $= \frac{\text{Stress}}{Y} = \frac{4\times10^6}{2\times10^{11}} = 2\times10^{-5}$

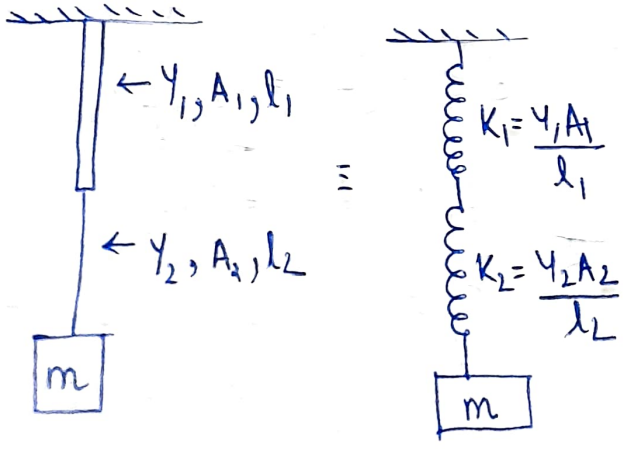
③ $\Delta l = \text{Strain} \times l = 2\times10^{-5}\times 2m = 4\times10^{-5}\ m$

④ Energy Stored $= \frac{1}{2}\times\text{Stress}\times\text{Strain}\times\text{Volume}$

$U = \frac{1}{2}\times4\times10^6\times2\times10^{-5}\times25\times10^{-6}\times2$

$U = 2\times10^{-3}$ Joules

Q



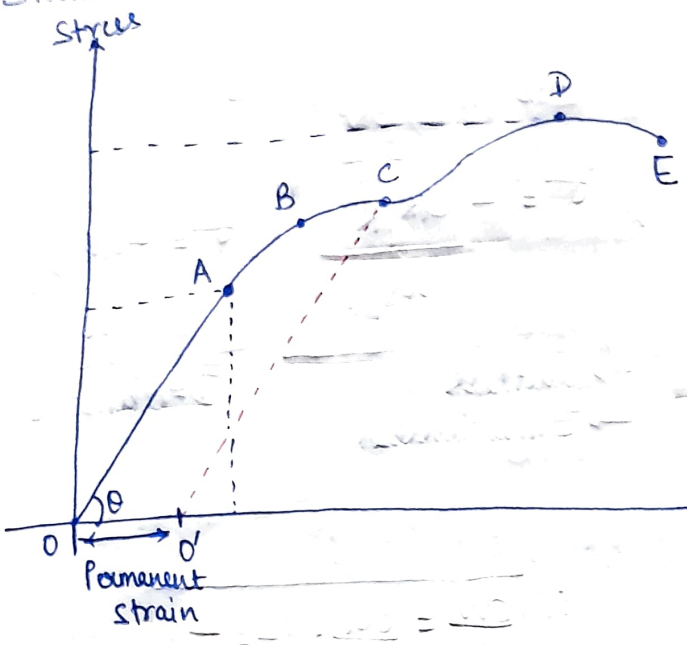
Find time period of oscillations of block.

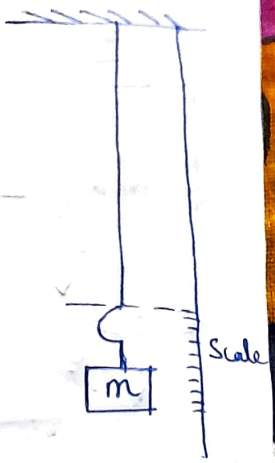
$$K_{eq} = \frac{K_1K_2}{K_1+K_2}$$

$$T = 2\pi\sqrt{\frac{m}{K_{eq}}} = 2\pi\sqrt{\frac{m(K_1+K_2)}{K_1K_2}}$$

$$T = 2\pi\sqrt{\frac{m\left(\frac{Y_1A_1}{l_1}+\frac{Y_2A_2}{l_2}\right)}{\frac{Y_1A_1}{l_1}\times\frac{Y_2A_2}{l_2}}}$$

Stress - Strain Curve:





① OA ⇒ Region of proportionality

⇒ Stress ∝ Strain

Point A = Proportional limit

$$\boxed{\text{Stress} = Y \text{ strain (Hookes law)}}$$

$$Y = \frac{\text{stress}}{\text{strain}} = \tan\theta$$

② Point B is elastic limit.

In region AB, Hooke's law is not valid. Small changes in stress can cause larger change in strain.

Till point B, wire have elastic property.

If we remove deforming force, wire will achieve, its actual shape, size and geometry.

Region OB is elastic region.

③ Point C is yield point. At this point even if we reduce stress, there will be some strain and now wire will have a permanent strain.

④ Point D is point of max^m stress / Breaking stress.

After reaching point D, if we even decrease stress, strain will increase, length of wire will keep on increasing, wire will get thinner, starts to flow and behaves like fluid.

⑤ Point E is fracture point. At this point wire breaks

Region BE is plastic region.

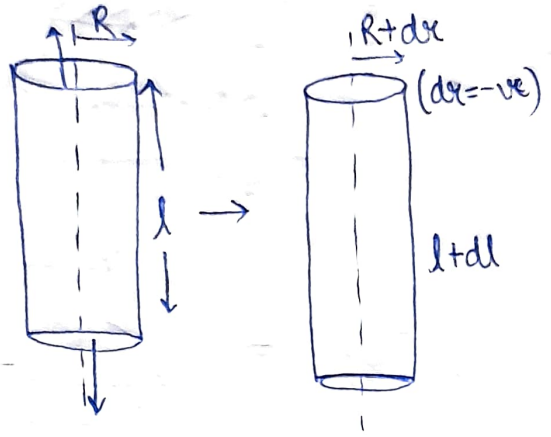
Note:

① Tensile/Ductile material ⇒ large plastic region

② Brittle material ⇒ B and E points are near ⇒ smaller plastic region.

As we cross point B, brittle material breaks.

Poisson's Ratio:



Poisson's Ratio = σ

$$\sigma = -\frac{dR/R}{dl/l} \qquad \sigma = \frac{-dD/D}{dl/l}$$

→ unitless
→ Dimensionless

D = diameter

Now, $V = \pi R^2 l$

Take ln both sides

$\ln V = \ln\pi + 2\ln R + \ln l$

Differentiating both sides

$$\frac{dV}{V} = 0 + 2\frac{dR}{R} + \frac{dl}{l}$$

$$\frac{dV}{V} = -2\sigma\frac{dl}{l} + \frac{dl}{l}$$

$$\therefore \boxed{\frac{dV}{V} = \frac{dl}{l}\{1-2\sigma\}}$$

% change in volume =

$$\frac{dV}{V}\times 100 = \frac{dl}{l}(1-2\sigma)\times 100$$

Note: ① If $\sigma = 1/2 \Rightarrow l\uparrow$ or $l\downarrow \Rightarrow V = fix$

② If $\sigma > 1/2 \Rightarrow l \propto \frac{1}{V}$ ③ If $\sigma < 1/2 \Rightarrow l \propto V$

Q A wire having poisson's ratio $\sigma = \frac{1}{4}$, is stretched to increase length by 1%. Find ① % change in radius ② % change in volume

① $$\sigma = \frac{-dR/R}{dl/l}$$

$$\frac{dR}{R}\times 100 = -\sigma\frac{dl}{l}\times 100$$

$$\frac{dR}{R}\times 100 = -\frac{1}{4}\times 1\%$$

$$\therefore \boxed{\frac{dR}{R}\times 100 = -0.25\%}$$

② $$\frac{dV}{V}\times 100 = \frac{dl}{l}(1-2\sigma)\times 100$$

$$\frac{dV}{V}\times 100 = 1\%\left(1-2\left(\frac{1}{4}\right)\right)$$

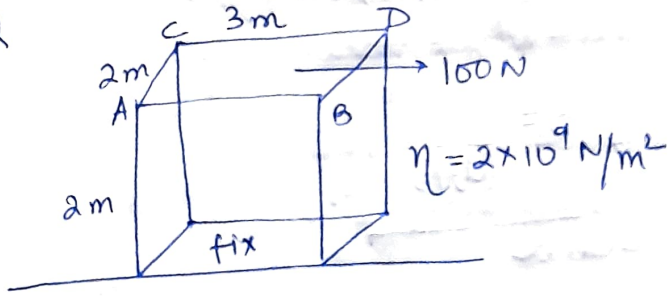
$$\boxed{\frac{dV}{V}\times 100 = 0.5\%}$$

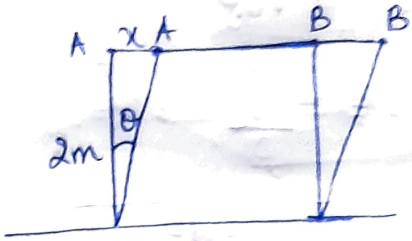
Modulus of Rigidity (η):

$$\eta = \frac{\text{Shear stress}}{\text{Shear strain}}$$

→ Unit = N/m^2
→ Dimension = $[ML^{-1}T^{-2}]$
→ Property of material

Q





Find: ① Tangential Stress $= \frac{F_t}{A} = \frac{100N}{6m^2} = \frac{50}{3} N/m^2$

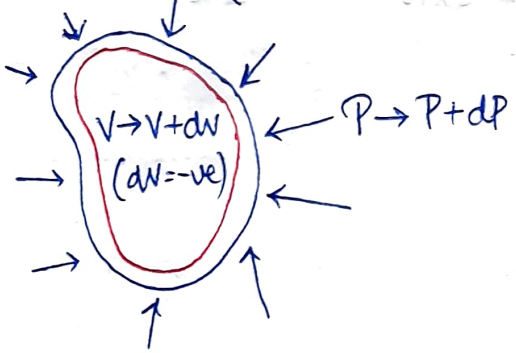
② Tangential Strain $= \frac{\text{Shear stress}}{\eta} = \frac{50}{3\times2\times10^9} = \frac{25}{3}\times10^{-9}$

③ Find x and θ.

Shear strain $= \frac{x}{h} \Rightarrow \frac{25}{3}\times10^{-9} = \frac{x}{2} \Rightarrow x = \frac{5}{3}\times10^{-8} m$

$\theta = \frac{25}{3}\times10^{-9}$ Radian

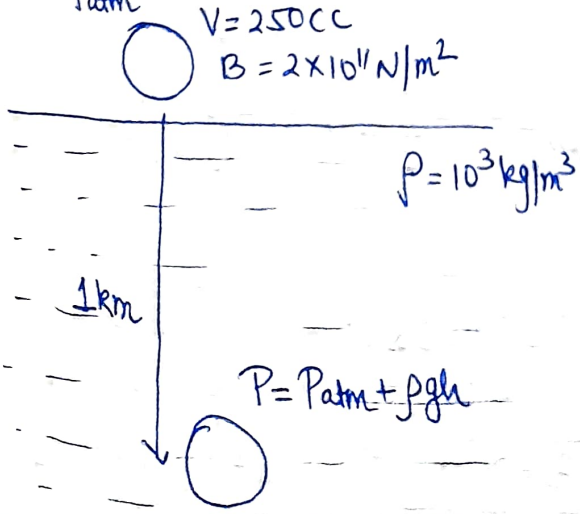
Bulk Modulus (B):



$$B = \frac{\text{Vol. Stress}}{\text{Vol. Strain}} = \frac{-dP}{dV/V}$$

$$\text{Compressibility} = \frac{1}{B}$$

Q



Find volume of ball at depth 1km

$\Delta P = \rho g h = 10^3 \times 10 \times 10^3 = 10^7 N/m^3$

$B = \frac{-10^7}{dV/250cc} \Rightarrow dV = \frac{-10^7 \times 250}{2\times10^{11}}$

$dV = -125\times10^{-4} cc$

$V_{final} = V + dV = 250 - 125\times10^{-4} cc$

Alternate formula of Bulk modulus:

$\rho V = m = \text{const}$

Differentiate

$\rho dV + V d\rho = 0$

$\rho dV = -V d\rho$

$\frac{dV}{V} = -\frac{d\rho}{\rho}$

$$\boxed{\frac{dV}{V} \times 100 = -\frac{d\rho}{\rho} \times 100}$$

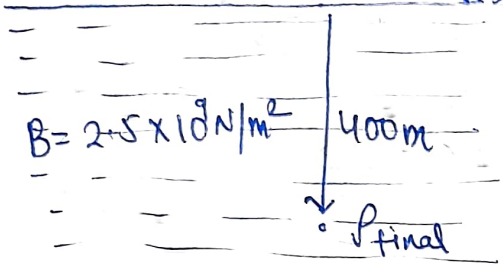
* for small changes,

If $V \uparrow \Rightarrow \rho \downarrow$

So if $V \uparrow$ by 1% $\Rightarrow \rho \downarrow$ by 1%

* $$\boxed{B = \frac{dP}{(d\rho/\rho)}}$$

Q



Find approximate value of ρ_{final}.

$dP = \rho g h = 10^3 \times 10 \times 400 = 4 \times 10^6 N/m^2$

$B = \frac{dP}{d\rho/\rho} \Rightarrow d\rho = \frac{dP}{B} \times \rho$

$d\rho = \frac{4 \times 10^6}{2.5 \times 10^9} \times 10^3 = \frac{4}{2.5} = 1.6 kg/m^3$

$\rho_{final} = (1000 + 1.6) kg/m^3$

Bulk modulus of gases:

→ Depends on process

① Isochoric Expansion ($V = \text{const} \Rightarrow dV = 0$)

$B = -\frac{dP}{dV/V} \longrightarrow \infty$ (not defined)

② Isobaric Expansion ($P = \text{const} \Rightarrow dP = 0$): $B = 0$

③ Isothermal Expansion:

$PV = \text{const}$

$P dV + V dP = 0$

$-\frac{dV}{V} = \frac{-dP}{P} \Rightarrow \boxed{\frac{dP}{dV} = -\frac{P}{V}}$

$B = -\frac{dP}{dV/V} = -\frac{dP}{dV} V = -\left(-\frac{P}{V}\right)(V)$

$\boxed{B = P}$

④ Polytropic Process:

$PV^a = \text{const}$

$\boxed{B = aP}$

⑤ Adiabatic Process:

$PV^\gamma = \text{const}$

$\boxed{B = \gamma P}$